

Aging Narratives in Canadian Media

Ascend.io Hackathon Project

Complete Implementation Guide

Generated: January 31, 2026

Hackathon Period: January 29 - February 6, 2026

Updated: Corrected Ascend UI terminology

This document contains step-by-step instructions for:

- Storing datasets in Azure Blob Storage
- Ingesting data into Ascend with Otto prompts
- Building ONE unified Flow/Pipeline for data cleaning

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SECTION 1: STORE DATASETS TO AZURE BLOB STORAGE

Estimated Time: 30-45 minutes

Task 1.1: Create Azure Account

1. Go to <https://azure.microsoft.com/en-us/free/>
2. Sign up with your email (free tier includes storage)
3. No credit card required for free tier

Task 1.2: Create a Storage Account

1. Log into Azure Portal (<portal.azure.com>)
2. Search for "Storage accounts"
3. Click "+ Create"
4. Fill in the following information:

Resource Group	Create new → Name it "aging-narratives-rg"
Storage Account Name	aging-narratives-storage (must be unique, lowercase)
Region	Canada Central (or closest to you)
Redundancy	Locally-redundant storage (LRS) - cheapest option

5. Click "Review + Create" → "Create"
6. Wait 2-3 minutes for deployment to complete

Task 1.3: Create Containers for Your Datasets

1. Go to your storage account (Storage Accounts → aging-narratives-storage)
2. Click "Containers" in left menu
3. Click "+ Container"
4. Create **Container 1**:
 - Name: **raw-articles-2023**
 - Public access level: Private

5. Create **Container 2**:

- Name: **raw-articles-2025-2026**
- Public access level: Private

Task 1.4: Get Connection Credentials

1. In Storage Account, click "Access Keys" (left menu)
2. Copy and save these in a safe place:
 - **Storage account name**: aging-narratives-storage
 - **Key1**: (copy the full key)
 - **Connection string**: (copy the full connection string)
3. You will need these for Ascend configuration

Task 1.5: Upload Your Datasets

1. Go to Containers → **raw-articles-2023**
2. Click "Upload"
3. Select your 2023 CSV file
4. Wait for upload to complete (progress bar shows status)
5. Repeat for **raw-articles-2025-2026** container with your 2025-2026 data

✓ Verification Check:

- Both containers should now contain your CSV files
- Files should show in the container with upload date/time

SECTION 2: SET UP ASCEND WORKSPACE & CONNECTION

Estimated Time: 15-20 minutes

Important Terminology Note:

In Ascend, a "Flow" is the same as a "Pipeline". They're synonymous terms.

Task 2.1: Log into Your Ascend Workspace

1. Go to <https://app.ascend.io>
2. Log into your workspace (you should already have access)
3. You should see your workspace listed (e.g., "Suha")
4. Click on your workspace to open it

Task 2.2: Create Azure Blob Storage Connection

1. In your workspace, locate the **"Connections"** section on the left sidebar
2. Click the "+" button or right-click to create a new connection
3. Select **"Azure Blob Storage"** from the connection type options
4. Name it: **azure-aging-narratives**
5. Enter your Azure credentials from Task 1.4:
 - **Storage Account Name:** aging-narratives-storage
 - **Storage Account Key:** (paste from Task 1.4)
6. Click "Test Connection" → should show ✓ Success
7. Save the connection

✓ Verification Check:

- Connection appears in your Connections list
- Status shows as "Connected" or ✓

SECTION 3: CREATE ONE FLOW/PIPELINE WITH ALL COMPONENTS

Estimated Time: 5 minutes to initialize, then 60 minutes to build components

Important Architecture Decision:

You will create **ONE Flow/Pipeline** that contains **ALL components**. This is better than creating multiple flows because:

- Cleaner, more professional data engineering architecture
- Easier to schedule and orchestrate (one schedule for everything)
- Shows judges you understand unified pipeline design
- Natural data flow: both datasets combine and flow through transforms together

Task 3.1: Create Your First (and Only) Flow/Pipeline

1. In your workspace, click the **"Create Flow"** button
2. Name it: **aging_narratives_pipeline**
3. This creates your main pipeline that will contain all 7 components
4. You should now be in the Flow editor

Task 3.2: Ask Otto to Set Up Both Read Connectors

Use this prompt with Otto:

I want to build a unified data pipeline in this flow. Can you help me set up TWO Read Connectors:

1. First Read Connector:
 - Name: read_articles_2023
 - Connect to: Azure Blob Storage using the 'azure-aging-narratives' connection
 - Read from: raw-articles-2023 container
 - File format: CSV
 - Columns: url, title, publish_date, media_name, phrase, phrase_category, province, article_text
 - Auto-detect schema
2. Second Read Connector:
 - Name: read_articles_2025_2026
 - Connect to: Azure Blob Storage using the 'azure-aging-narratives' connection
 - Read from: raw-articles-2025-2026 container
 - File format: CSV
 - Same columns as above

- Auto-detect schema

Please show me both connectors in the flow diagram, and provide a preview of data from each connector to verify they're working correctly.

This is all within ONE flow called 'aging_narratives_pipeline'.

Expected Otto Output:

- Two Read Connectors created with specified names
- Both showing as connected to Azure Blob Storage
- Data previews from both containers
- Visual flow diagram showing both components

✓ **Verification Check:**

- Both connectors show 'Success'
- Data previews show your article columns
- Row counts visible from both containers

SECTION 4: TRANSFORM COMPONENT 1 - CLEANING & PREPROCESSING

Estimated Time: 60-90 minutes

Overview:

You will now add 5 Transform components to your ONE flow. These transforms will clean both datasets and combine them.

Task 4.1: Understand Data Quality Issues

Use this prompt with Otto:

```
I'm about to add Transform components to this flow to clean my datasets. The data has these columns: url, title, publish_date, media_name, phrase, phrase_category, province, article_text.
```

```
Can you help me identify common data quality issues I should clean:
```

1. What null/missing value problems might exist?
2. What date format issues should I watch for?
3. What text cleaning is needed (whitespace, encoding, special characters)?
4. What deduplication logic should I apply (by URL)?
5. What province name standardization might be needed?

```
The provinces are: Ontario, Ottawa, BC, Quebec
```

Task 4.2: Create First Transform - Clean 2023 Data

Use this prompt with Otto to add Transform Component 1:

```
Add a Transform component to the flow that cleans the 2023 data. Connect it to the 'read_articles_2023' connector.
```

```
This transform should:
```

1. ****Remove Duplicates****:
 - Deduplicate by URL (keep first occurrence)
 - Remove rows where URL is NULL
2. ****Standardize Province Names****:
 - Map variations to standard values: 'Ontario', 'Ottawa', 'British Columbia', 'Quebec'
 - If province doesn't match, mark as 'Other'
 - Create column 'province_clean'
3. ****Clean Text Fields****:
 - Trim whitespace from title, phrase, article_text
 - Convert phrase_category to lowercase

- Handle NULL values in article_text with empty string ''
4. ****Date Standardization****:
 - Standardize publish_date to YYYY-MM-DD format
 - Handle unparseable dates by setting to NULL
 - Create 'year' column from publish_date
 - Create 'month' column from publish_date
 5. ****Add Metadata****:
 - Add 'ingestion_date' column (today's date)
 - Add 'dataset_source' = '2023_media_cloud'
 - Add 'row_hash' as unique identifier

Name this transform output: 'cleaned_articles_2023'

Show me: preview of cleaned data + row count before/after

Task 4.3: Create Second Transform - Clean 2025-2026 Data

Use this prompt with Otto to add Transform Component 2:

Add another Transform component to clean the 2025-2026 data.
Connect it to the 'read_articles_2025_2026' connector.

Apply the EXACT same cleaning transformations:

1. Remove duplicates by URL
2. Standardize province names (Ontario, Ottawa, British Columbia, Quebec)
3. Trim whitespace from text fields
4. Standardize phrase_category to lowercase
5. Standardize publish_date to YYYY-MM-DD format
6. Extract year and month columns
7. Add ingestion_date (today's date)
8. Add dataset_source = '2025_2026_media_cloud'
9. Add row_hash as unique identifier

Name the output: 'cleaned_articles_2025_2026'

Show me: preview of cleaned data + row count before/after + any NULL values

Task 4.4: Create Third Transform - Combine Both Datasets

Use this prompt with Otto to add Transform Component 3:

Add a Transform component that combines both cleaned datasets using UNION.

Connect it to BOTH:

- 'cleaned_articles_2023'
- 'cleaned_articles_2025_2026'

This transform should:

1. Combine both datasets using UNION ALL
2. Ensure column order is consistent across both
3. Add a 'dataset_period' column:
 - '2023' for rows from cleaned_articles_2023
 - '2025_2026' for rows from cleaned_articles_2025_2026

Name the output: 'all_cleaned_articles'

Show me:

- Total row count for combined dataset
- Distribution of articles by year and province

Task 4.5: Create Fourth Transform - Add Data Quality Checks

Use this prompt with Otto to add Transform Component 4:

Add a final Transform component for data quality checks.
Connect it to 'all_cleaned_articles'.

This component should check and report:

1. ****NULL value checks**** in critical columns:
 - url (should be 0 NULLs)
 - title (acceptable: < 5% NULLs)
 - publish_date (should be 0 NULLs)
 - province_clean (should be 0 NULLs)
 - phrase_category (should be 0 NULLs)
2. ****Value validation****:
 - phrase_category should only contain: 'limiting', 'neutral', 'empowering'
 - province_clean should only contain: 'Ontario', 'Ottawa', 'British Columbia', 'Quebec', 'Other'
3. ****Generate a quality summary showing****:
 - Total rows processed
 - Rows with NULLs by column
 - Any invalid values found
 - Data quality score

Name the output: 'quality_checked_articles'

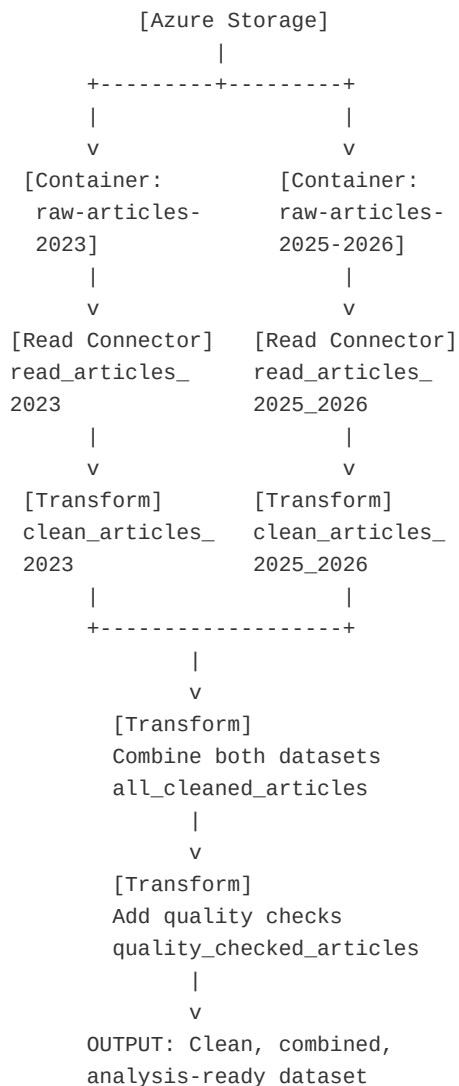
This is the final, analysis-ready dataset.

✓ Verification Check After All Transforms:

- All 4 transform components appear in flow diagram
- Data flows smoothly from connectors → transforms → final output
- Final output 'quality_checked_articles' is clean and analysis-ready
- No errors in any component

SECTION 5: YOUR COMPLETE PIPELINE ARCHITECTURE

Single Flow: aging_narratives_pipeline



Pipeline Summary:

- ✓ **Flows:** 1 unified flow/pipeline
- ✓ **Read Connectors:** 2 (one for each dataset)
- ✓ **Transform Components:** 4
- ✓ **Total Components:** 6 (exceeds 4-component minimum)

- ✓ **Data Flow:** Both datasets ingested, cleaned separately, combined, quality-checked
- ✓ **Ready for:** Analysis, aggregation, visualization, and weekly orchestration

SECTION 6: EXECUTION TIMELINE & BREAKDOWN

Day 1	Task Duration	Activities
Morning	1-1.5 hours	Tasks 1.1 - 1.5: Azure setup + upload datasets
Mid-Morning	15-20 min	Tasks 2.1 - 2.2: Ascend workspace + Azure connection
Noon	5 min	Task 3.1: Create ONE flow/pipeline
Afternoon	20-30 min	Task 3.2: Add both Read Connectors to flow
Late Afternoon	1-1.5 hours	Task 4.1 - 4.2: Add first Transform (clean 2023)
Day 2	Task Duration	Activities
Morning	30-45 min	Task 4.3: Add second Transform (clean 2025-2026)
Mid-Morning	20-30 min	Task 4.4: Add Union Transform (combine datasets)
Afternoon	30-45 min	Task 4.5: Add Quality Check Transform
Late Afternoon	30 min	Verify entire flow runs end-to-end
End of Day 2	5-6 hours total	COMPLETE: Analysis-ready dataset in ONE pipeline

Total Time Investment: 5-7 hours over 2 days

Remaining Time for Next Phases:

- Days 3-5 (estimated 12-15 hours): Add analysis transforms (aggregation, temporal trends, etc.)
- Day 6 (estimated 2-3 hours): Set up orchestration/scheduling for weekly updates
- Day 7 (estimated 3-4 hours): Write submission + record demo video

SECTION 7: KEY TIPS FOR SUCCESS

1. One Flow/Pipeline is Better

You're using **ONE Flow** instead of multiple flows. This is the correct approach because:

- Shows professional data engineering thinking
- Makes orchestration simpler (one schedule)
- Natural data combination and processing flow
- Judges appreciate unified architecture

2. Use Otto Conversationally

If Otto's output isn't quite right, refine with follow-up prompts:

- "Can you add a column for..."
- "That's close, but can you also include..."
- "The output should also show..."

3. Test After Each Component

Don't assume everything works end-to-end on the first try:

- Run each Read Connector individually first
- Test each Transform separately before connecting to the next
- Check row counts at each stage to spot data loss issues

4. Screenshot Everything

For your hackathon submission, screenshots are crucial evidence:

- Screenshots of Otto building components (shows agentic features in action)
- Complete flow diagram showing all components connected
- Data preview/output samples from each stage
- Quality check results

5. Handle Province Names Carefully

Note: Ottawa is a city, not a province. Check your data:

- If "Ottawa" should map to "Ontario", update the standardization
- If "Ottawa" is genuinely separate, keep it in the lookup
- Use consistent standardization across all transforms

6. Save All Prompts

Copy all prompts from this document into a text file so you can:

- Reference them if you need to rebuild components
- Modify them for future iterations
- Share with team members if collaborating

7. Run the Complete Flow

Before moving to next phases, run your entire flow end-to-end:

- All 6 components should execute successfully
- No errors or warnings
- Final 'quality_checked_articles' output should be clean and ready

8. Document Your Decisions

For your final submission writeup, explain:

- Why you chose a unified flow architecture
- How you used Otto to accelerate development
- Data quality issues found and how they were resolved
- Specific cleaning/standardization decisions made

QUICK REFERENCE CHECKLIST

Azure Setup Checklist

- Created Azure account with free tier
- Created Storage Account: aging-narratives-storage
- Created Container: raw-articles-2023
- Created Container: raw-articles-2025-2026
- Uploaded 2023 CSV to raw-articles-2023
- Uploaded 2025-2026 CSV to raw-articles-2025-2026
- Saved Access Keys & Connection string

Ascend Setup Checklist

- Logged into Ascend workspace
- Created Azure Blob Storage connection: azure-aging-narratives
- Connection test shows Success

ONE Flow/Pipeline Checklist

- Created ONE flow named: aging_narratives_pipeline
- Added Read Connector: read_articles_2023
- Added Read Connector: read_articles_2025_2026
- Added Transform: clean_articles_2023
- Added Transform: clean_articles_2025_2026
- Added Transform: all_cleaned_articles (UNION)
- Added Transform: quality_checked_articles
- All 6 components visible in flow diagram

Flow Execution Checklist

- Ran complete flow end-to-end successfully
- No errors or warnings in any component

- All row counts match expectations
- Final output is clean and analysis-ready
- Took screenshots of complete flow diagram

SUPPORT RESOURCES & NEXT STEPS

During the Hackathon:

Ascend Community Slack: Join the #agentic-data-hackathon channel

Link: ascend.io/slack

Office Hours: Get live help from Ascend team

Dates: February 3 & 5, 2026 at 9am PT

Register: https://us02web.zoom.us/meeting/register/fanxmASPSQGZIWdLR-_HSQ#/registration

Documentation: Technical guides and tutorials

Link: docs.ascend.io

What's Next (After Completing This Section):

Days 3-5: Add Analysis Transforms (12-15 hours)

- Transform 5: Aggregate data by province, phrase category, time period
- Transform 6: Calculate sentiment ratios (limiting vs. empowering)
- Transform 7: Temporal trend analysis (articles over time)

Day 6: Set Up Orchestration (2-3 hours)

- Schedule your flow to run weekly
- Configure to automatically detect new articles
- Set up notifications for failures

Day 7: Finalize Submission (3-4 hours)

- Write 500-800 word submission
- Record 5-minute demo video showing complete pipeline
- Submit via: <https://docs.google.com/forms/d/e/1FAIpQLSdB29MkYRu3P5jGbxqI3dy1qCAC1ybGBdVmFGpJTsZ0XNpujQ/viewform>

Hackathon Submission Deadline:

Friday, February 6, 2026 at 11:59 PM PST

What to Include in Your Submission:

- ✓ Brief writeup (500-800 words) describing your solution
- ✓ How you used Ascend's agentic features (Otto, etc.)
- ✓ How you set up the unified pipeline architecture
- ✓ Data quality issues found and resolved
- ✓ Screenshots or 5-minute demo video of your ONE pipeline

Good luck with your hackathon project! ■